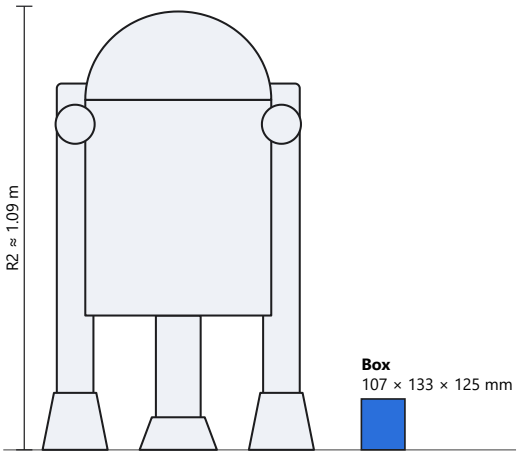
ROS2 box · Assembly instructions

Mini-tower box for the Raspberry Pi 4, the Pico board, the 24 → 5 V converter, the XT60 and the switch. Before printing, measure what the README lists (XT60, switch, converter holes, button and fan heights).

Up-to-date guide, STL and sources:
https://github.com/BeskarCode/ROS2D2/tree/main/hardware/caja_ros2

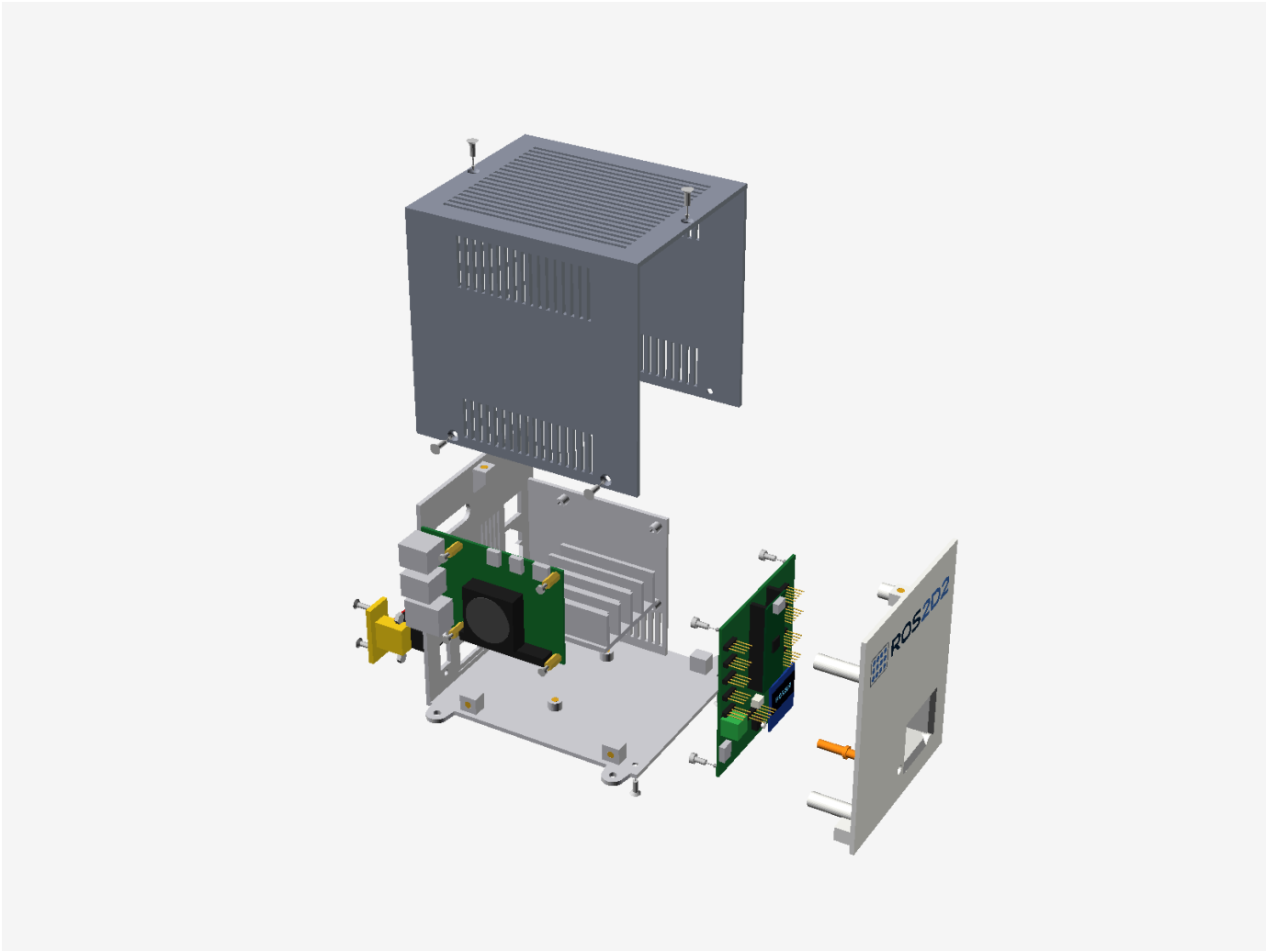




Size next to the R2

Schematic R2 and the box at the same scale.

Exploded view



Hardware and parts

Printed parts

`frontal_multicolor_bambu.3mf`	Front with the logo in 4 colours (AMS: white, navy, blue, grey)
`frontal_monocolor.stl`	or: single-colour front, engraved logo
`base.stl`	Base: floor, back, Pi bracket
`carcasa.stl`	U shell (roof + sides)
`pulsador.stl`	DISPLAY push rod

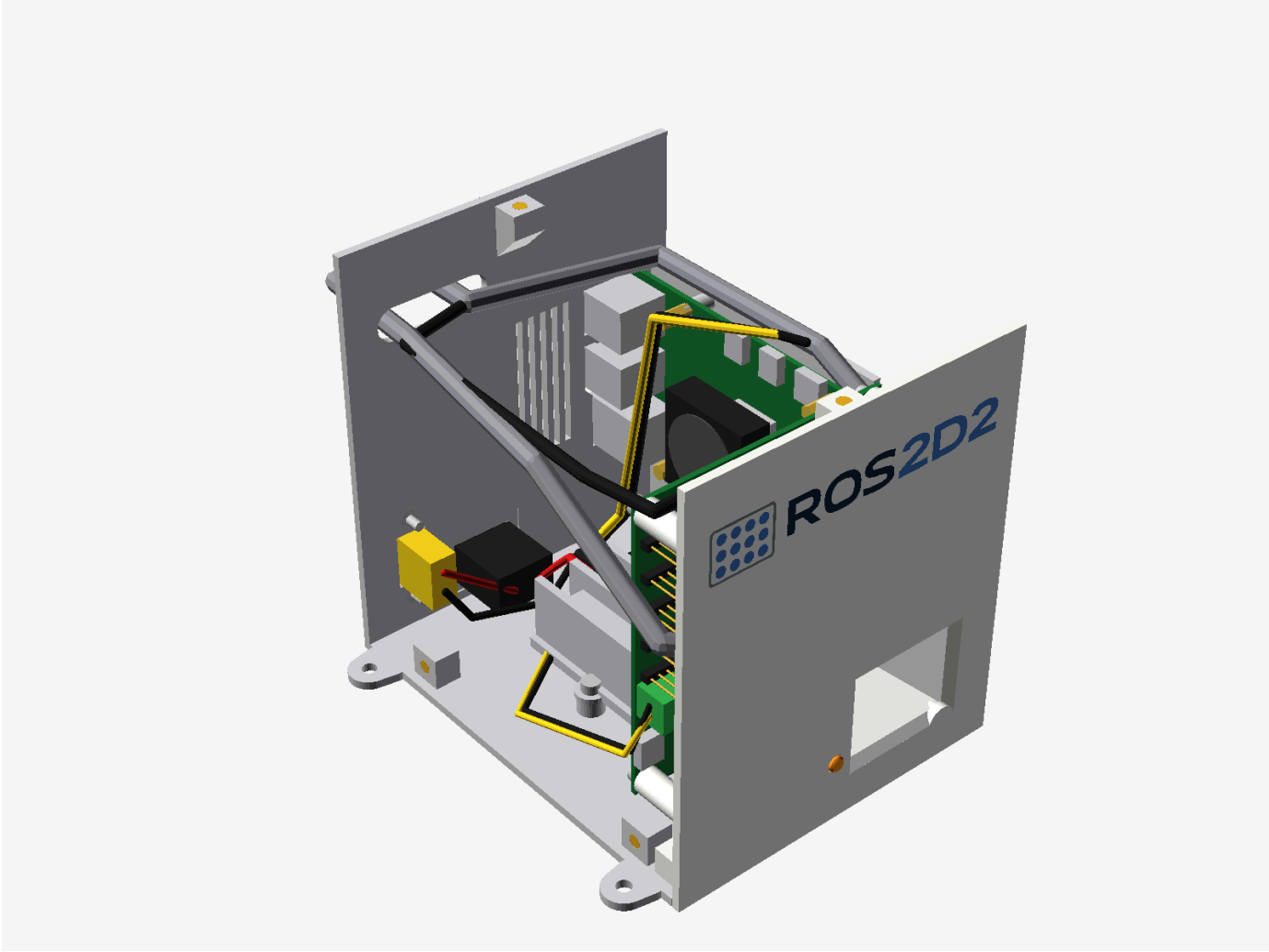
Qty	Part	Step
14×	M3 heat-set insert (5.7 mm long, 4.0 mm hole)	1
8×	M3 × 8 countersunk screw (ISO 10642)	6, 7
4×	M3 × 6 button head screw (ISO 7380)	5
2×	M3 × 6 socket head cap screw (DIN 912)	2
2×	M3 × 10 countersunk screw (ISO 10642)	3
2×	M3 nut (DIN 934)	3

4×	M2.5 × 6 screw (longer if the Pi's fan sits on spacers)	4
4×	M4 × 12 screw (or what the R2's mount needs)	8

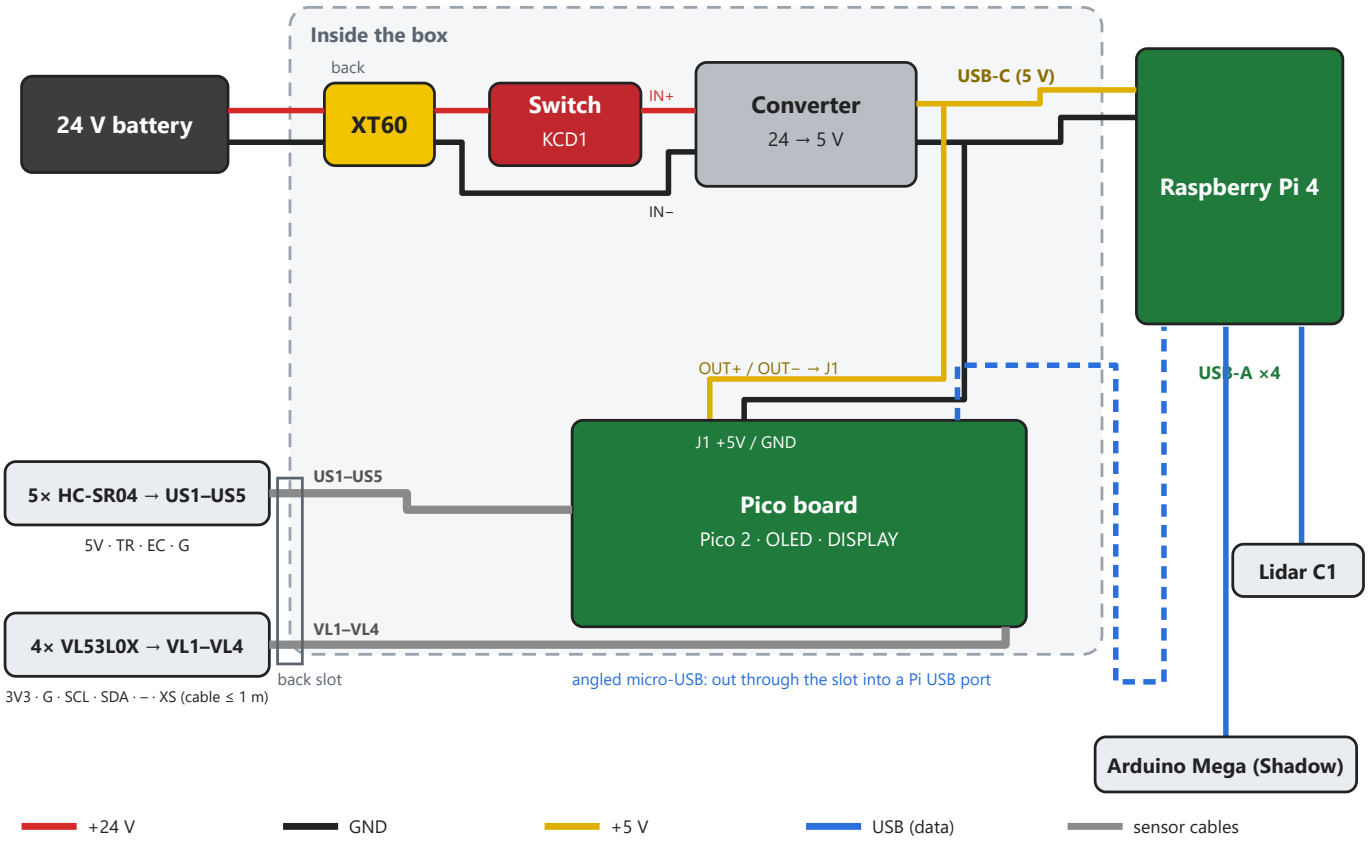
Other parts

1	KCD1 rocker switch (panel hole 19.2 × 13 mm)
1	XT60E-F panel connector (AMASS)
1	micro-USB 2.0 to USB-A cable, 90° side angle, 30 cm (see «Angled plugs»)
1	USB-C male pigtail (5 V to the Pi), 3 A, 15–20 cm; straight or 90° angled
—	0.5 mm^2 wire for the 5 V to terminal J1, 1 mm^2 for the 24 V
1	Edge grommet strip for the 35 × 10 mm slot

Inside (no shell)



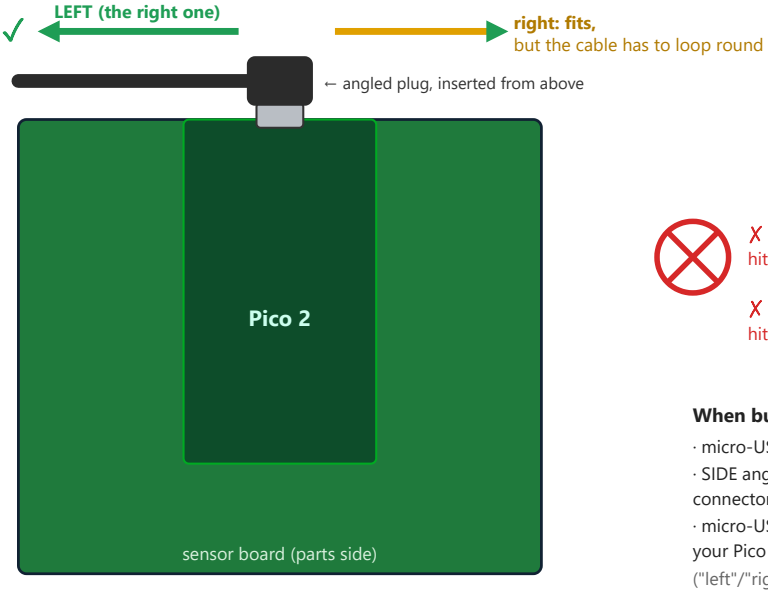
Wiring



Angled plugs: which way the cable goes

1 · Angled micro-USB (Pico → Pi) · both boxes

Looking at the Pico's parts side, its USB at the top (as it sits on the sensor board):



✗ **towards you (the parts side):**
hits the sensors' Dupont plugs

✗ **backwards (behind the board):**
hits the divider in box C

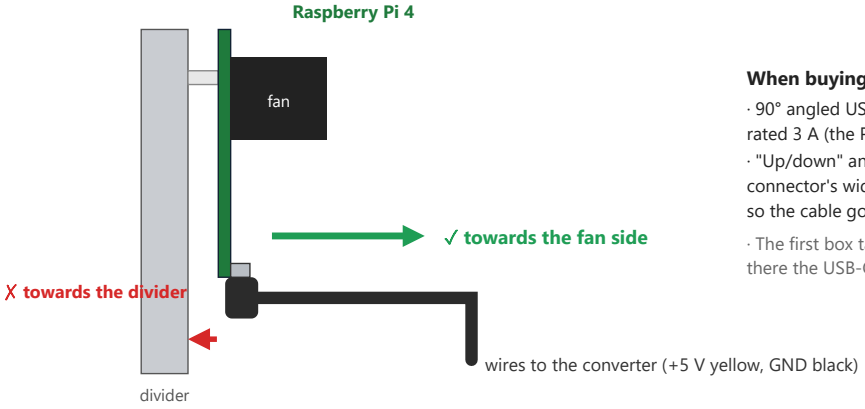
When buying:

· micro-USB 2.0 (not 3.0), 30 cm (fits both boxes).
· SIDE angle: the cable leaves sideways, along the connector's long side (not up or down).
· micro-USB only goes in one way: compare the photo with your Pico and check the cable will end up on the LEFT.
("left"/"right" mean different things to different brands)

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2 · Angled USB-C pigtail (converter → Pi) · box C

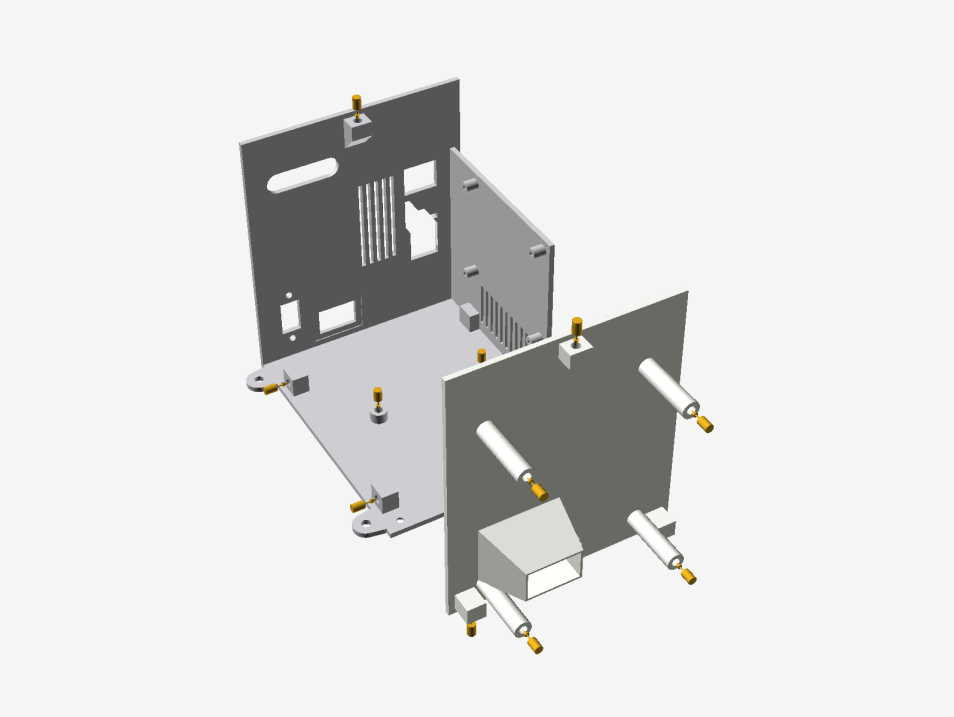
The Pi seen from the narrow front: standing, parts and fan to the right, USB-C at the bottom:



When buying:

· 90° angled USB-C male pigtail (loose wires), rated 3 A (the Pi 4 wants 5 V / 3 A), 15–20 cm.
· "Up/down" angle: the cable leaves square to the connector's wide face. USB-C is reversible: turn it so the cable goes towards the fan.
· The first box takes the same cable (or a straight one): there the USB-C faces the roof, cable towards the middle.

1 Heat-set inserts

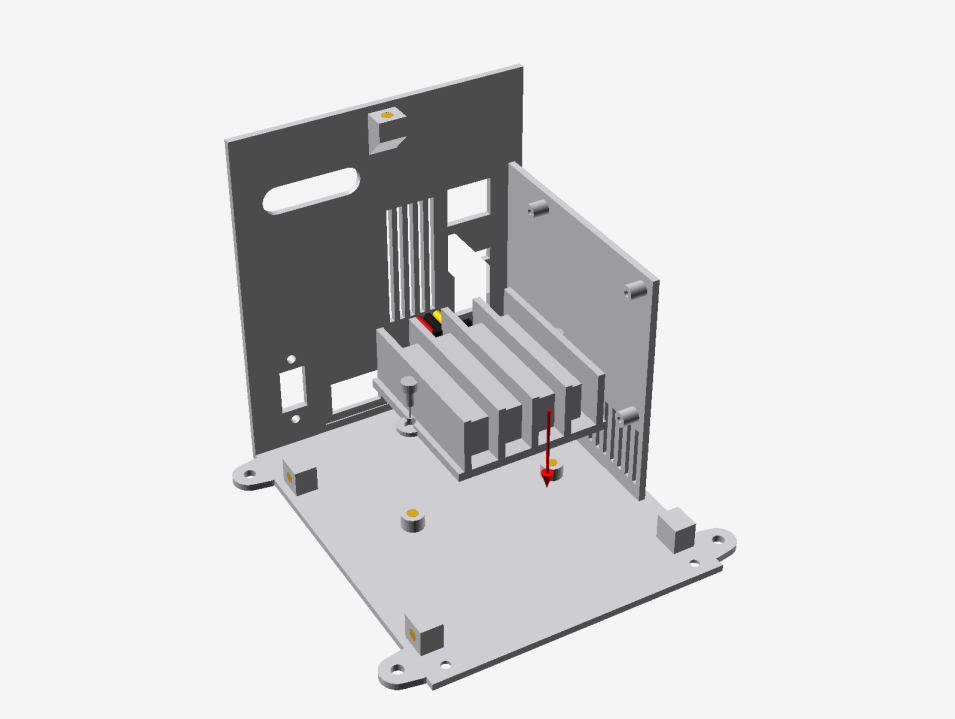


14× M3 insert

With the soldering iron at about 220 °C, press the inserts in straight and flush: on the front, 4 in the columns, 2 in the bottom tabs and 1 in the top tab; on the base, 4 in the floor blocks, 2 in the converter bosses and 1 in the top tab.

2

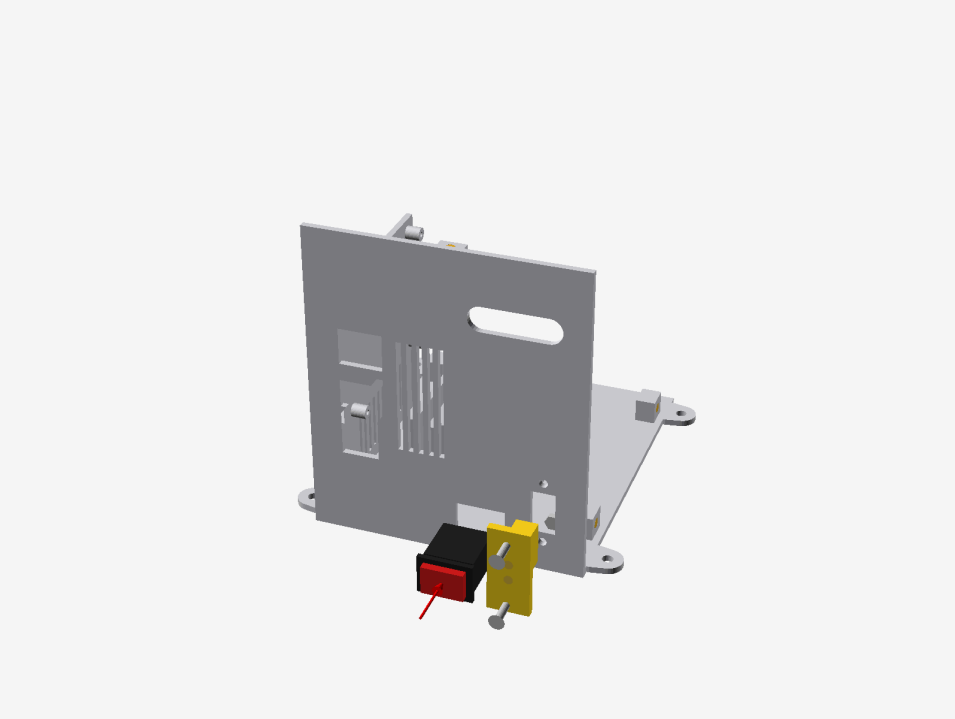
24 → 5 V converter



2× M3 × 6 socket head

Screw it onto its two bosses, wires towards the back. Check its holes (`conv_holes`) with a caliper before printing the base.

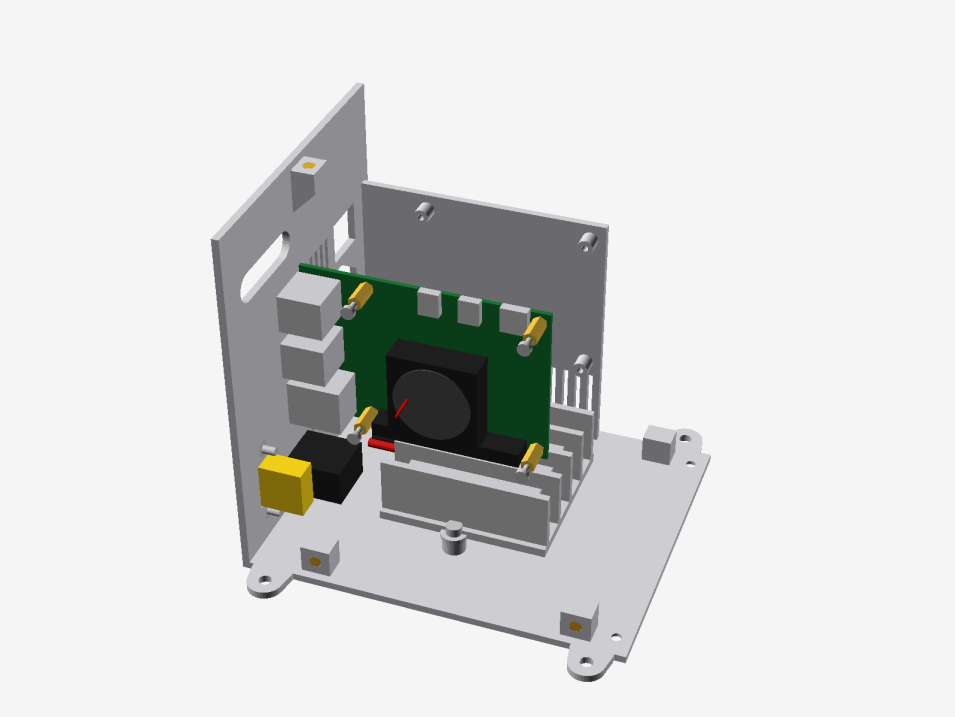
3 XT60 and switch



2× M3 × 10 countersunk | 2× M3 nut

From the outside: the XT60E-F with 2 screws and their nuts inside, and the switch snapped in. Wire the 24 V: XT60 → switch → converter input.

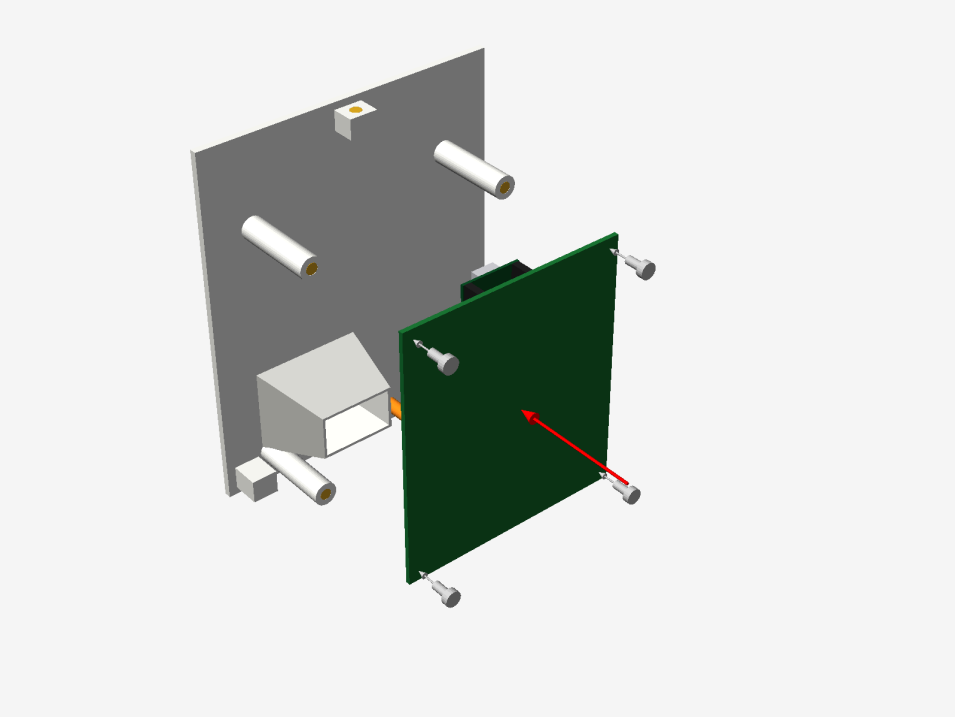
4 Raspberry Pi



4× M2.5 × 6

Hold it against the bracket a few mm forward and slide it back until the USB and Ethernet jacks come flush out of the back. Screw it on and plug in the 5 V USB-C from the converter.

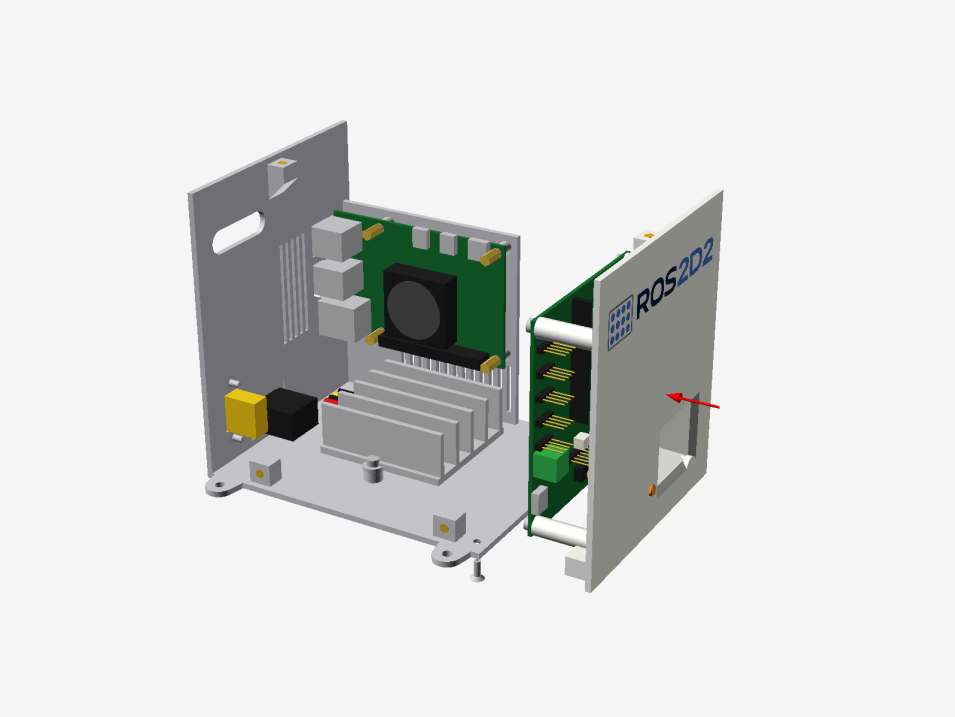
5 Push rod and Pico board



4× M3 × 6

Slide the push rod into its guide from the inside. Screw the board onto the 4 columns, parts facing the front. With the front still loose: tighten the 5 V in terminal J1, plug the angled micro-USB into the Pico and connect the sensors.

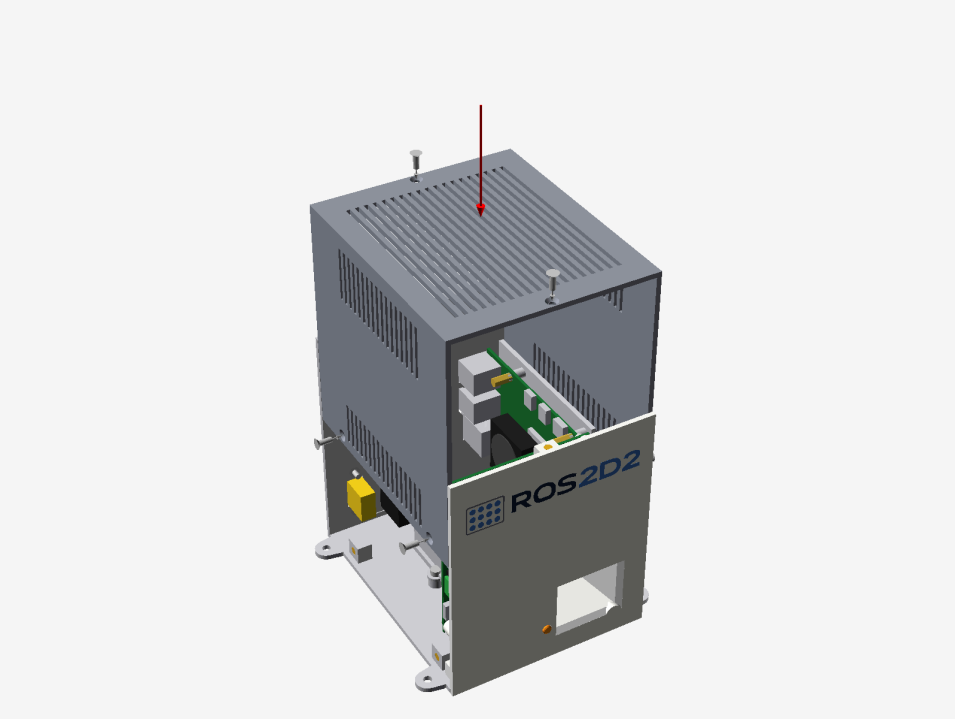
6 Front onto the base



2× M3 × 8 countersunk

Run the sensor cables and the micro-USB along the channels beside the board to the slot in the back. Screw the front on from under the floor.

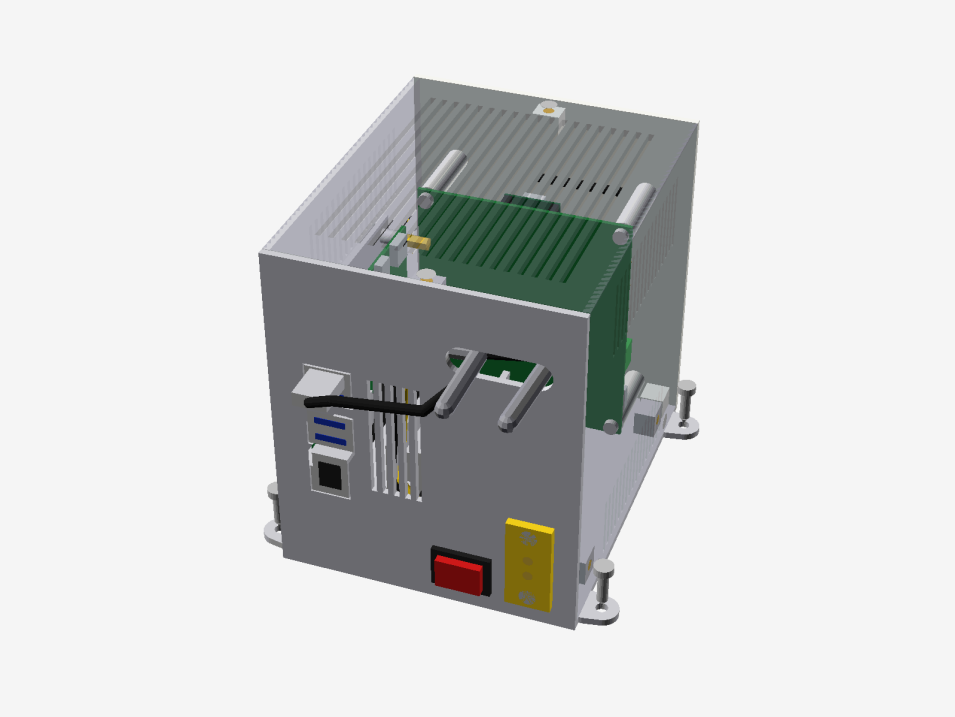
7 Shell



6× M3 × 8 countersunk

Lower it from above and fit the 4 side screws and the 2 roof screws.

8 USB and onto the R2



4× M4 × 12

Bring the micro-USB out through the slot into a Pi USB port. Screw the box to the R2 through its 4 ears (use a hex key: they are 6 mm from the wall).